

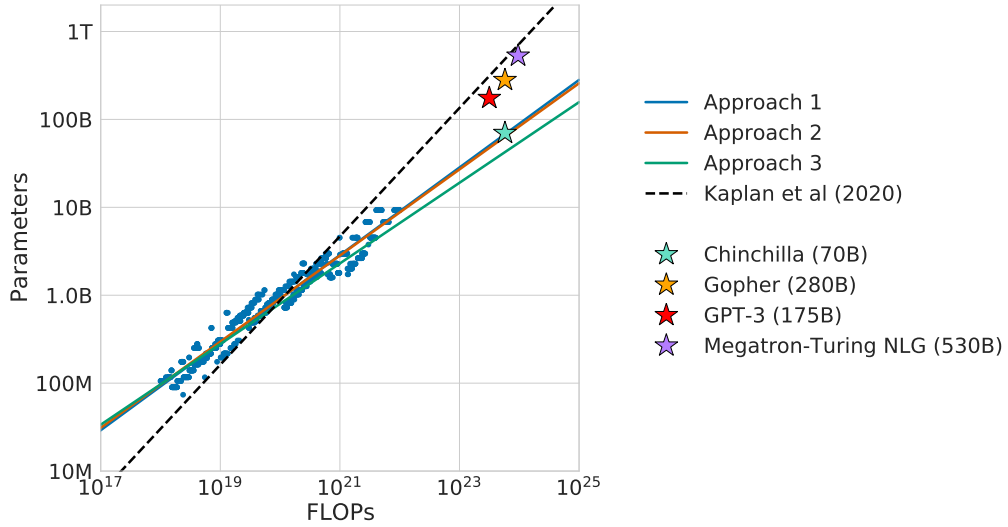


Figure 1 | [Kaplan et al. \(2020\)](#) $\log$ - $\log$ plot of Parameters vs. FLOPs, showing the scaling of different models. The x-axis represents FLOPs (ranging from 10^{17} to 10^{25}) and the y-axis represents Parameters (ranging from 10M to 1T). The plot includes three solid lines representing different scaling approaches (Approach 1, Approach 2, Approach 3) and a dashed line representing the scaling from Kaplan et al. (2020). Data points for Chinchilla (70B), Gopher (280B), GPT-3 (175B), and Megatron-Turing NLG (530B) are shown as stars.

Given N Tokens and D Tokens, the FLOPs of the model is $FLOPs(N, D)$, and the FLOPs of the model is $FLOPs(N, D) = C$ for L :

$$N_{opt}(C), D_{opt}(C) = \underset{N, D \text{ s.t. } FLOPs(N, D) = C}{\operatorname{argmin}} L(N, D). \quad (1)$$

For $N_{opt}(C)$ and $D_{opt}(C)$ for C ranging from 70M to 16B, 5B to 400B Tokens, we follow the scaling of [Kaplan et al. \(2020\)](#) in [Figure 1](#) and [Section 2](#) to estimate the optimal scaling.

For example, for $C = 4$, the optimal scaling is 4 Tokens and 4 Tokens, which is 70B Tokens, and 1.4 Tokens for *Gopher*, which is 280B Tokens, and 175B Tokens for *GPT-3*, and 530B Tokens for *Megatron-Turing NLG*.

2. Scaling

Transformer models (Brown et al., 2020; Lieber et al., 2021; Rae et al., 2021; Smith et al., 2022; Thoppilan et al., 2022) and MoE models (Du et al., 2021; Fedus et al., 2021; Zoph et al., 2022) are Transformer models with 5000 layers (Smith et al., 2022) and are trained on a large dataset. The scaling of Transformer models is different from the scaling of MoE models. The scaling of Transformer models is different from the scaling of MoE models. The scaling of Transformer models is different from the scaling of MoE models.

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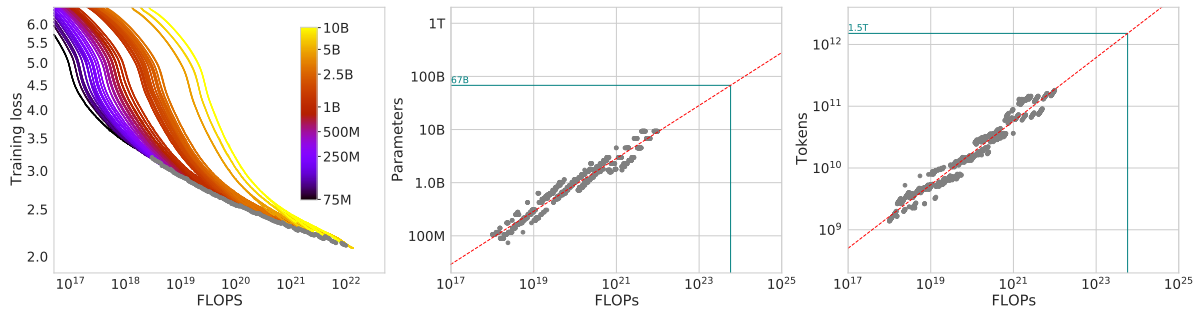


Figure 2 | 70M 10B, FLOP , Token (5.76 × 10²³) Token

Clark et al. (2022) , Kaplan et al. (2020) ,

Transformer , McCandlish et al. (2018) , Shallue et al. (2018); Zhang et al. (2019) , Levine et al. (2020)

Transformer , MoE 1.7 Switch Transformer (Fedus et al., 2021) 1.2 GLaM (Du et al., 2021) (Artetxe et al., 2021; Zoph et al., 2022) FLOP, (Clark et al., 2022) Transformer, (Borgeaud et al. (2021); Guu et al. (2020); Lewis et al. (2020) Token (Borgeaud et al. (2021) ~ 10

3. Token

FLOPs , Token ? , Clark et al. (2022); Kaplan et al. (2020) , Token ³ Table 2

3.1. 1: Token

70M 10B , 4 FLOP

N , 4 , Token 10× , 16× , FLOP , FLOP C N Token D , $FLOPs(N, D) = C \cdot D^4$ 1500 FLOP , Token (Figure 2

³ FLOPs

⁴ 15% D Token , D Token 10×

Table 3 | FLOP and Token counts, where 1 FLOP and Token are 2 or 3 bytes (???)

	FLOPs	FLOPs (B)	Tokens
4 B	$1.92\text{e}+19$	1/29,968	80 B
10 B	$1.21\text{e}+20$	1/4,761	202 B
100 B	$1.23\text{e}+22$	1/46	2051 B
670 B	$5.76\text{e}+23$	1	1.5 GB
1750 B	$3.85\text{e}+24$	6.7	3.7 GB
2800 B	$9.90\text{e}+24$	17.2	5.9 GB
5200 B	$3.43\text{e}+25$	59.5	11.0 GB
1 GB	$1.27\text{e}+26$	221.3	21.2 GB
10 GB	$1.30\text{e}+28$	22515.9	216.2 GB

However, IsoFLOP is not a good metric for LLM, as it does not consider the Token counts.

???, IsoFLOP is not a good metric for LLM, as it does not consider the Token counts. (Raffel et al., 2020) and GitHub (Rae et al., 2021) are not good, as they do not consider the Token counts.

4. Chinchilla

Section 3, 400 to 700 Token counts, 700 Token counts, 1.4 Token counts, Chinchilla, Token counts

(Rae et al., 2021) 4x,

4.1. Chinchilla

Table 4, Chinchilla

- MassiveText (B) Chinchilla, Chinchilla Token counts
- (Loshchilov and Hutter, 2019) Adam (Kingma and Ba, 2014), Chinchilla⁸
- SentencePiece (Kudo and Richardson, 2018) tokenizer Chinchilla, tokenizer B NFKC Chinchilla—94.15% Token counts
- bfloat16 (Rajbhandari et al., 2020) float32 Chinchilla, Rae et al. (2021)

???, TPuv3/TPuv4 (Jouppi et al., 2017) JAX (Bradbury et al., 2018) Haiku (Hennigan et al., 2020) (Mitchell et al., 2019)

	Key/Value	d_{model}	Token	Token
Gopher 280B	80 128	128	16,384	4×10^{-5} 3M $\rightarrow$ 6M
Chinchilla 70B	80 64	128	8,192	1×10^{-4} 1.5M $\rightarrow$ 3M

Table 4 | key/value and d_{model} (Token) $4 \times d_{\text{model}}$, Chinchilla

⁸Chinchilla, AdamW 80% Adam Chinchilla, Chinchilla

	Count	Examples
WikiText-103	20	WikiText-103, The Pile: PG-19, arXiv, FreeLaw, ...
RACE-m, RACE-h, LAMBADA	3	RACE-m, RACE-h, LAMBADA
Natural Questions, TriviaQA, TruthfulQA	3	Natural Questions, TriviaQA, TruthfulQA
HellaSwag, Winogrande, PIQA, SIQA, BoolQ	5	HellaSwag, Winogrande, PIQA, SIQA, BoolQ
MMLU	57	... , ...
BIG-bench	62	... , ...

Table 5 | The Chinchilla paper (Rae et al. (2021)) shows that the Chinchilla model is the most efficient, achieving the highest performance per FLOP across a wide range of tasks.

4.2. Data

The Chinchilla paper (Rae et al. (2021)) shows that the Chinchilla model is the most efficient, achieving the highest performance per FLOP across a wide range of tasks. This is achieved by using a large dataset of 1.6 trillion tokens, which is the largest dataset used for training a language model at the time of writing.

4.2.1. The Pile

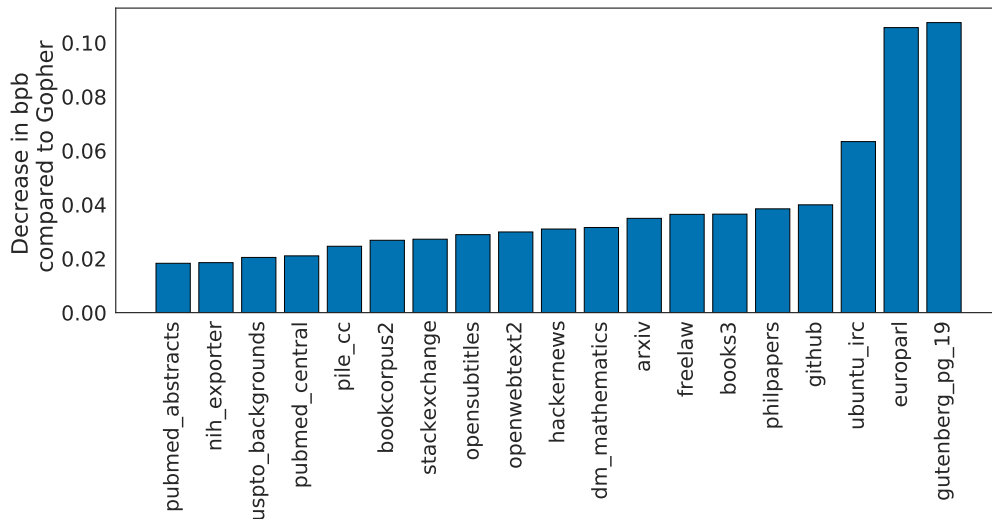


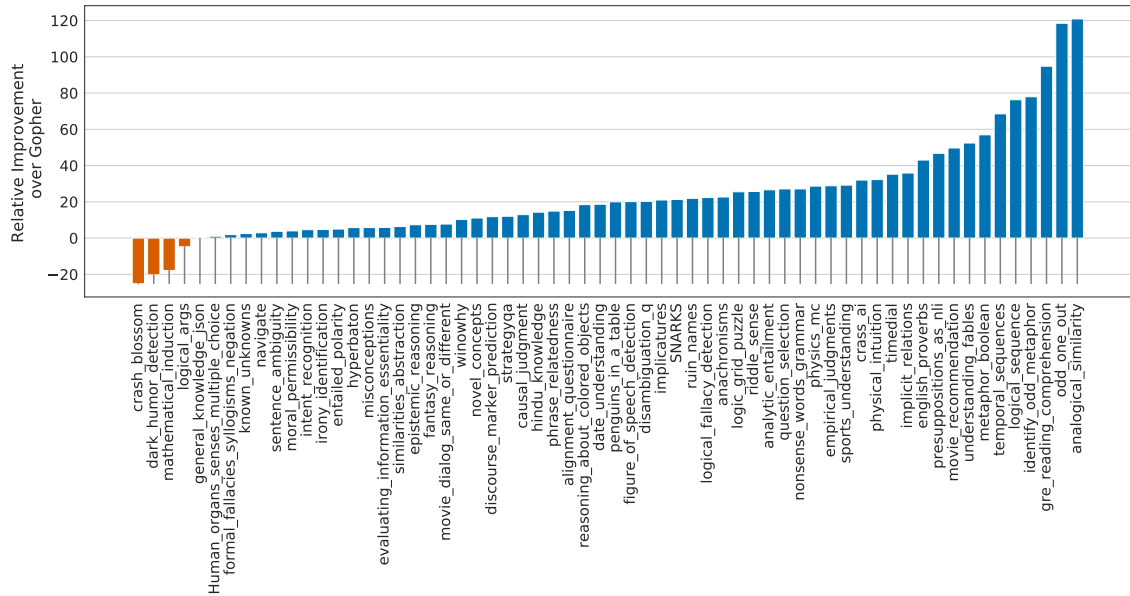
Figure 5 | The Pile (Gao et al., 2020) is a large dataset of 1.6 trillion tokens, which is the largest dataset used for training a language model at the time of writing. The chart shows the decrease in bpb compared to Gopher for various datasets.

The Pile (Gao et al., 2020) is a large dataset of 1.6 trillion tokens, which is the largest dataset used for training a language model at the time of writing. The chart shows the decrease in bpb compared to Gopher for various datasets. The datasets are sorted by the decrease in bpb, with europarl and gutenber_pg_19 having the highest decrease in bpb.

4.2.2. MMLU

The MMLU (Hendrycks et al., 2020) is a dataset of 150 multiple-choice questions across 57 subjects. The chart shows the performance of the Chinchilla model on MMLU, which is 67.6% (7.6% improvement over the previous state-of-the-art). The chart also shows the performance of the Chinchilla model on other datasets, such as high_school_gov_and_politics, international_law, sociology, and us_foreign_policy, which all achieve 90% accuracy.

Figure 6 shows the performance of the Chinchilla model on other datasets, such as college_mathematics, econometrics, and moral_scenarios, which all achieve 90% accuracy.



	<i>Chinchilla</i>	<i>Gopher</i>	GPT-3	MT-NLG 530B
LAMBADA □□□	77.4	74.5	76.2	76.6
RACE-m □□□	86.8	75.1	58.1	-
RACE-h □□□	82.3	71.6	46.8	47.9

4.2.5. □□

	<i>Chinchilla</i>	<i>Gopher</i>	GPT-3	MT-NLG 530B	□□ SOTA
HellaSWAG	80.8%	79.2%	78.9%	80.2%	93.9%
PIQA	81.8%	81.8%	81.0%	82.0%	90.1%
Winogrande	74.9%	70.1%	70.2%	73.0%	91.3%
SIQA	51.3%	50.6%	-	-	83.2%
BoolQ	83.7%	79.3%	60.5%	78.2%	91.4%

	□□	<i>Chinchilla</i>	<i>Gopher</i>	GPT-3	SOTA (□□)
Natural Questions (dev)	0-shot	16.6%	10.1%	14.6%	54.4%
	5-shot	31.5%	24.5%	-	
	64-shot	35.5%	28.2%	29.9%	
TriviaQA (□□□, test)	0-shot	67.0%	52.8%	64.3 %	-
	5-shot	73.2%	63.6%	-	
	64-shot	72.3%	61.3%	71.2%	
TriviaQA (□□, dev)	0-shot	55.4%	43.5%	-	72.5%
	5-shot	64.1%	57.0%	-	
	64-shot	64.6%	57.2%	-	

Table 9 | □□□□□□ Natural Questions (Kwiatkowski et al., 2019) □ TriviaQA (Joshi et al., 2017), *Gopher* □□ Natural Questions □, -3□□ TriviaQA □, □□□□□□□□□□□□□□, □□□ GPT-3 □□□ SOTA(FiD + Distillation (Izacard and Grave, 2020)) □□□□□

	<i>Chinchilla</i>	<i>Gopher</i>		<i>Chinchilla</i>	<i>Gopher</i>
□□	78.3%	71.4%	□□ <i>gotcha</i>	62.5%	59.2%
□□	71.2%	68.0%	□□ <i>not gotcha</i>	80.0%	76.7%
□□	79.6%	71.3%	□□ <i>gotcha</i>	76.7%	66.7%
□□	84.2%	75.0%	□□ <i>not gotcha</i>	82.5%	75.8%

Table 10 | Winogender 0000: 00: (gotcha 00) 00000000, 0000000000

4.2.6. □□□□

Table 9 Natural Questions (Kwiatkowski et al., 2019) □, :5-shot □ 31.5%,64-shot □ 35.5%, □□□□ 21% □ 28%□ TriviaQA (Joshi et al., 2017) □, □□□□□□□□ (□□□□□□□□□□) □□□□ (□□□□□□□□□□) □□□□□□□□□□,*Gopher* □□□□□□□□,Chinchilla □□□□□ SOTA (Izcard and Grave, 2020) 7.9%□□□□ □□□,-3——□Table 9□

4.2.7. □□□□□□□

000000000000, 0000000000000000000000 (Bender et al., 2021; Weidinger et al., 2021)00000, 00, 00
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[illegible]

Table 10 shows, for each category (at 3.2%), the number of items (at 8.3% to 9.2%) that are *gotcha*, the number of items that are not *gotcha* (at 10.0%), the number of items that are *gotcha/not gotcha* (at 10.0%), the number of items that are *gotcha* (at 10.0%), and, for each category, the number of items that are not *gotcha*.

00000 000000000000——0000000000000000 (Gehman et al., 2020; Rae et al., 2021)0000000000, 00
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